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Session 7 — AI Integration for the 1-Week Revision Plan

Objective

Connect the application to an AI API and use module data to generate a personalized 7-day revision plan for the student.

1. Why AI in This Project?

The AI is not just a decoration — it solves a real problem. Given the student's modules, their difficulty, progress, understanding, and exam dates, the AI can generate a smart weekly study schedule. This is the core intelligence of your application.

The AI will use these fields to prioritize:

Technology	What It Does
difficulty	Harder modules get more study time
career_importance	Important modules are prioritized
progress	Low progress = more time needed
understanding_level	Low understanding = more urgent review
exam_date	Nearest exams are scheduled first

2. Where to Get an AI API?

Option 1: GitHub Models / GitHub Marketplace

Beginner-friendly for developers. Good if you already use GitHub. Try different AI models easily.

Option 2: Hugging Face

Access many free and open models. Great for experimentation. Popular in AI student projects.

Option 3: OpenRouter (alternative)

One API key gives access to multiple AI models. Practical for simple integration without managing multiple accounts.

Recommendation: Choose the API with the clearest documentation. Start with a simple text generation endpoint. Do not pick a complicated setup for this project.

3. How to Build the AI Prompt

Send your module data to the AI as a structured prompt. Example:

```
You are a study planner. Create a 7-day revision plan for:
- Web Techniques: Difficulty=Hard, Progress=40%, Exam=in 5 days
- Mathematics: Difficulty=Medium, Progress=70%, Exam=in 10 days
Prioritize urgent and difficult modules. Format: Day 1, Day 2, ...
```

4. Practical Tasks This Week

1	Register for an API key at GitHub Models, Hugging Face, or OpenRouter
2	Create generate_plan.php with a button to generate the revision plan
3	Fetch all modules of the logged-in user from the database
4	Build a structured text prompt from the module data
5	Send the prompt to the AI API using PHP (curl or file_get_contents)
6	Display the returned text as a 7-day revision plan on the page
7	Optional: save the generated plan in the study_plans table

Expected Result by End of Week

- ✓ The student can click a button to generate a personalized revision plan
- ✓ The AI uses the actual module data (difficulty, progress, exam dates)
- ✓ The plan is displayed clearly on the page
- ✓ The AI feature is genuinely useful and not just decorative

Advice & Common Mistakes to Avoid

- ! Make sure your full CRUD system works perfectly before starting the AI integration
- ! Start by testing the API manually (with a fixed prompt) before connecting it to real data
- ! Keep the prompt short and structured — clear instructions give better AI results
- ! Store your API key in a separate config file and never expose it publicly
- ! If the API fails, display a helpful error message to the user

Prepare for Next Week

- → Test every feature end-to-end: registration, login, CRUD, and AI plan
- → Fix all remaining bugs and improve the interface
- → Prepare a clear demo scenario for the final presentation